

Ayushman Singh

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EDUCATION

Sardar Vallabhbhai National Institute of Technology

Surat, Gujarat

Bachelor of Technology in Computer Science and Engineering

August 2024 – Present

- Current CGPA: 8.67/10.0
- Relevant Coursework: Data Structures, Design and Analysis of Algorithms, Database Management System, Object Oriented Programming, Computer Network, Artificial Intelligence, Operating System

EXPERIENCE

Software Development Intern

May 2026 - Jul 2026

SchooliP Learning

Remote

- Developed and shipped a **multi-tenant ERP and Learning Management System** using **Next.js, Firebase, and Firestore**, serving administrators, faculty, students, and parents across multiple schools.
- Engineered production-ready modules for **attendance, examinations, syllabus management, leave workflows, analytics, and role-based access control**, contributing to core platform functionality.
- Optimized Firestore data access through **query redesign and multi-level caching**, reducing database reads by **84%** while improving dashboard responsiveness and application performance.
- Implemented **offline-first Progressive Web App (PWA)** capabilities, collaborated on debugging, testing, deployment, and delivered responsive user experiences across desktop and mobile devices.

Summer Research Intern

May 2025 - Jun 2025

MNNIT Allahabad, Prayagraj

On-site

- Developed a real-time **person detection and tracking pipeline** using **YOLO, OpenCV, and NumPy** for dense crowd surveillance as part of Mahakumbh 2025 public-safety research.
- Built a **Kalman Filter-based tracking pipeline** with appearance-aware re-identification to maintain identities across occlusions and reduce identity switches in crowded environments.
- Designed and evaluated modular detection, tracking, and re-identification components to improve robustness under challenging visibility, motion, and heavy crowd conditions.

PROJECTS

JoyJuncture | *Next.js, Firebase, Firestore*

Website | GitHub

- Built a full-stack **gamified community platform** integrating e-commerce, event management, blogging, and reward-based engagement using **Next.js, Firebase, and Firestore**.
- Designed backend workflows for authentication, wallet management, bookings, payments, and Firestore-backed data storage across **15+ collections**, ensuring secure and scalable application logic.
- Integrated **Razorpay, Google Maps, Google Calendar, Cloudinary, and chatbot automation**, while developing an admin dashboard supporting analytics, moderation, and complete CRUD operations.

TransitFlow | *React, Node.js, Express.js, MongoDB*

Website | GitHub

- Developed a modular urban mobility platform integrating **smart parking, carpooling, public transit information, and multi-modal route planning** using **React, Node.js, Express.js, and MongoDB**.
- Designed RESTful APIs and MongoDB data models for ride management, parking availability, and transportation workflows with a scalable backend architecture.
- Implemented interactive route visualization and fare comparison across multiple transport modes, improving accessibility and enabling users to compare alternative transportation options efficiently.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, C, HTML/CSS, SQL

Frameworks: React, Node.js, Express.js, Next.js, Tailwind CSS

ML / Data: Pandas, NumPy, PyTorch

Computer Vision: YOLO, OpenCV

Databases: MongoDB, MySQL, Firestore

Tools: Git, GitHub, Postman, REST APIs

ACHIEVEMENTS

- **Winner, Web Wonders 2025** for developing an innovative transportation solution.
- **1st Runner-Up, Enfinity Hackathon 2026** and **Runner-Up, Google Winter of Code 2025–26**.
- Solved **700+** Data Structures and Algorithms problems across competitive programming platforms.